

Ads Notes

YR

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1 Ranking Score

In ads ranking, the ranking score is often estimated as:

$$\text{Ranking Score} = \text{Bid} \times pCTR \times pCVR$$

which represents the expected value (e.g., expected revenue) per impression.

Auction: rank ads by score and determine which ads are shown and sometimes their prices. (展示广告的排序和位置)

User experiences will affect when considering pCTR.

(User experiences 会影响CTR, 但另一方面来说, 不能说pCTR直接受user experience影响, 因为“图片吸引人”或者“标题具有引导性”都会提成pCTR但是用户体验是negative的)

2 Commonly used feature engineering

- user features
 - historical CTR, demographics, user embedding
- ad features
 - creative(image, text, video features), ad history, ad category , historical performance
- Historical statistical features
 - user CTR over 1/7/30 days.
 - ad CTR by category , user-specific ad CTR
- cross features
 - user clicked the ads before? (user $\times$ ad /category /advertiser)
 - user-ad interaction frequency
- context features
 - time, device, position, hour of the day, connection type
- features crossing for nonlinear relationship
 - user age $\times$ ad category

3 常用训练模型

Increasing capability to model feature interaction

- Logistic Regression: Linear model, suitable for sparse features. Relay heavily on manual feature engineering
- LR+ feature crossing
- GBDT (Gradient Boosted Decision Trees): 优势: automatic feature partitioning, nonlinear interactions, no feature scaling.
常用: XGboost, LightBGM
- GBDT + LR

- Deep CTR models: DeepFM / DCN (deep cross network). for low and high order interactions
- DIN (deep Interest Network): Models users interests from **historical behaviour sequences** using attention.
eg: 点击历史, 浏览历史, 购买历史。
- DIEN/Transformer-based models

4 Cold Start

Modern approach: Content-based embeddings + representation learning

For new ads:

- content based features
 - categorical features: advertiser-ID, category, campaign type
 - creative features: image(handle with CNN or Vision Transformer), text embedding (ad title or description)
 - format features: video/ image, color, placement(广告位),...
- leverage similar ads
 - build ad embedding share the same or similar category /advertiser-ID
 - use KNN in embedding space
- exploration strategy
 - user Thompson Sampling or Upper Confidence Bound (UCB) to initially show the ad to small and representative sample
 - start with prior CTR (like category average value) and update
 - For first 1000 impression, for example, on "exploration", then on real data.

For new user

- contextual signals and request time
 - Device models, OS, network type. time of day, location, languages, traffic source(from website/app/events...)
- lookalike model
 - nearest cluster of existing user
- rapid adaptation: 新用户虽然一开始没有历史数据, 但只要产生少量行为, 就应该尽快利用这些行为更新模型对该用户的理解, 而不是一直依赖默认画像。
 - Continuously update user embeddings as interactions arrive.

Embeddings can be learned from historical click data using two-tower models, Word2Vec-style objectives, or deep CTR models.

5 案例分析: AUC improved but revenue dropped

Suppose AUC improved 2% but revenue dropped 1%. What will you do.

- Immediate check:
 - Data leaks?
 - verify feature distribution
 - check feature freshness
- Model behaviour analysis
 - calibration check
 - segment-level analysis:
 - * Ad category: over predicted low-bid ads
 - * User segment: how traffic allocated to different user segments? (high/low value users, new users, returning users)

- * traffic sources: IOS/Android, different country. Compared with trained data
- position bias
- Auction Dynamics
 - Model may over-predicted CTR for low-bid ads, and under-predicted CTR for high-bid ads. (训练时便宜的多, 贵的少, 导致实际收益下降)
 - * Label imbalance. Model calibration error. Objective mismatch. Distribution shift.
 - If in section-price auction, ranking = pCTR * bid.

	Bid	True CTR	pCTR	Score	Revenue
Ad-A	10	5%	4%	0.4	0.5
Ad-B	3	8%	15%	0.45	0.24

We see, Ad-B has a higher score, but actually it gains less revenue than Ad-A.

- Exploitation / reduced exploration. Reduced exploration may repeatedly expose similar ads, improving short-term CTR while hurting long-term revenue. It may show fewer diverse ads or fewer diverse users based on previous data. (short-term CTR improved, AUC improved, but these users were exposed to repeated/similar context, which led to low conversion rate.)
- Calibration: Poor calibration changes auction rankings even when AUC improves.

How to fix:

- Recalibrate the model using **isotonic regression or temperature scaling**
- add a value-aware or bid-weighted loss function. loss function to penalize errors from high-bid ads
- Implement position-aware training to account where ads appear
- AUC isn't everything. Monitor online metrics such as revenue, eCPM, CTR, CVR, and calibration in addition to AUC.
- Online/Offline distribution shift. Was the offline validation set consistent with online traffic?

6 Online Learning

6.1 Type A: Incremental Learning (True Online)

Update model parameters after each sample or mini-batch. Incremental learning updates the model continuously as new data arrives, rather than retraining the model from scratch.

Loss = cross-entropy(pred, Actual)

Model.weights -= learning rate × gradient

Pros: Pros: adapts quickly to distribution changes

cons: noisy, can forget old patterns.

6.1.1 复习: For multi-class classification.

For multi-class classification, the cross entropy loss is defined as:

$$L = - \sum_{k=1}^K y_k \log(p_k) \quad (1)$$

k: 类别索引index. K: 总类别数

eg:

True Label: [1,0,0]

Pred 1: [0.7, 0.2, 0.1]

Pred 2: [0.9, 0.05, 0.05]

$$L_1 = -[1 \times \log(0.7) + 0 + 0] = -\log(0.7) \approx 0.357$$

$$L_2 = [-1 \times \log(0.9) + 0 + 0] = -\log(0.9) \approx 0.105$$

Pred 2 assigns a higher probability to the correct class and therefore has a lower cross-entropy loss.
softmax function:

$$P_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}} \quad (2)$$

$$L = - \sum_{k=1}^K y_k \log\left(\frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}\right) \quad (3)$$

$$\frac{\partial L}{\partial z_k} = P_k - y_k \quad (4)$$

Binary cross entropy loss:

$$L_{total} = -\frac{1}{N} \sum_{n=1}^N [y_n \log(P_n) + (1 - y_n) \log(1 - P_n)] \quad (5)$$

对于CTR Prediction, Sigmoid- Binary Cross Entropy - Gradient. 因为广告点击预测是binary classification.

6.2 Type B: Streaming Retraining (Most common in industry)

Keep the model architecture fixed, but periodically reinitialize and retrain using a rolling window of recent data.

工业界常说: Periodic Retraining. Continuous Retraining. Rolling window Retraining.

eg: Take last 7 days of data - retrain model from scratch - deploy new model.

- Pros: Stable; adapts to recent data patterns, easy to monitor and deploy
- Cons: High computational cost and delayed adaptation to sudden distribution shifts. 今天发生的新趋势, 要下一轮训练才学到, 比如新闻热点。

Widely used because it is easier to monitor, validate, and safely deploy than true online learning.